

## 工 艺 规 程

工艺名称 J/RT148-02-02机加工工艺规程

产品型号 J/RT-148

零组件号 J/RT148-02-02

零组件名称 1区计量活门

总页数 24

初样研制

H 69-G0493-25 朱永江2025-02-28

设计版次

A

单 位

工艺版次

A01

军品机加分厂

一四三厂

编 号

69Y-3076

工艺

PDM 系统有效文件

|        |         |             |                |          |      |               |      |                 |               |        |  |
|--------|---------|-------------|----------------|----------|------|---------------|------|-----------------|---------------|--------|--|
| 一四三厂   |         | 综 合 目 录     |                |          |      | 编 号           |      | 69Y-3076        |               |        |  |
| 军品机加分厂 |         |             |                |          |      | 共 5 页         |      | 第 1 页           |               |        |  |
| 产品型号   |         | J/RT-148    |                | 零组件号     |      | J/RT148-02-02 |      | 零组件名称           |               | 1区计量活门 |  |
| 材料牌号   |         | 9Cr18       |                | 材料标准     |      | GJB 2294-95   |      | 毛坯类型            |               | 棒料     |  |
| 工 序    | 工 序 名 称 | 设 备         |                | 工 艺      |      | 装 备           |      | 备 注             |               |        |  |
|        |         | 名称          | 型号             | 名称       | 夹具图号 | 刀具图号          | 量具图号 |                 |               |        |  |
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| 5      | 调质      |             |                |          |      |               |      |                 | 特殊过           |        |  |
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| 10     | 车零件     | 数控车床        | CTX310V1       | 螺纹塞规     |      |               |      | 0420/M4-6H T    |               |        |  |
|        |         | 数控车床        | ELITE 42 ULTRA | 螺纹塞规     |      |               |      | 0420/M4-6H Z    |               |        |  |
|        |         | 数控车床        | ELITE 8/51     | 卡板       |      |               |      | 0340/2H12 TZ    |               |        |  |
|        |         | 车削中心        | QUEST 8/51     | 卡板       |      |               |      | 0340/1H13 TZ    |               |        |  |
|        |         |             |                |          |      |               |      |                 |               |        |  |
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| 初样研制   |         |             |                |          |      |               |      |                 |               |        |  |
| 会 签    | 15      | 车零件         | 数控车床           | CTX310V3 | 塞规   |               |      | 0220/5HJ12 T, Z |               |        |  |
|        |         |             |                |          |      |               |      | 编制              | 朱永江 2025/2/28 |        |  |
|        |         |             |                |          |      |               |      | 校对              | 胡文杰 2025/2/28 |        |  |
|        |         |             |                |          |      |               |      | 标审              | 袁贻乾 2025/2/28 |        |  |
|        |         |             |                |          |      |               |      | 审核              |               |        |  |
| 工艺版次   | H       | 69-G0493-25 | 朱永江2025-02-28  |          |      |               |      | 审定              |               |        |  |
| A01    | 标记      | 更改单号        | 签 名            | 标记       | 更改单号 | 签 名           | 批准   | 王建朋 2025/2/28   |               |        |  |

工艺

PDM 系统有效文件

|        |            |             |               |         |      |               |                  |               |  |        |  |
|--------|------------|-------------|---------------|---------|------|---------------|------------------|---------------|--|--------|--|
| 一四三厂   |            | 综 合 目 录     |               |         |      | 编 号           |                  | 69Y-3076      |  |        |  |
| 军品机加分厂 |            |             |               |         |      | 共 5 页         |                  | 第 2 页         |  |        |  |
| 产品型号   |            | J/RT-148    |               | 零组件号    |      | J/RT148-02-02 |                  | 零组件名称         |  | 1区计量活门 |  |
| 材料牌号   |            | 9Cr18       |               | 材料标准    |      | GJB 2294-95   |                  | 毛坯类型          |  | 棒料     |  |
| 工<br>序 | 工 序<br>名 称 | 设 备         |               | 工 艺 装 备 |      |               |                  | 备注            |  |        |  |
|        |            | 名称          | 型号            | 名称      | 夹具图号 | 刀具图号          | 量具图号             |               |  |        |  |
|        |            | 数控车床        | GS 200/66PLUS | 螺纹塞规    |      |               | 0421/M6-6H T , Z |               |  |        |  |
|        |            | 数控车床        | CTX310V1      |         |      |               |                  |               |  |        |  |
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| 20     | 铣槽         | 小型钻孔中心      | TC—S2Z        |         |      |               |                  |               |  |        |  |
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| 25     | 去毛刺        | 钳工台         |               |         |      |               |                  |               |  |        |  |
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| 30     | 标印         | 激光打标机       | YLP-HM30D     |         |      |               |                  |               |  |        |  |
| 初样研制   |            |             |               |         |      |               |                  |               |  |        |  |
| 会 签    |            |             |               |         |      |               |                  |               |  |        |  |
|        |            |             |               |         |      |               | 编制               | 朱永江 2025/2/28 |  |        |  |
|        |            |             |               |         |      |               | 校对               | 胡文杰 2025/2/28 |  |        |  |
|        |            |             |               |         |      |               | 标审               | 袁贻乾 2025/2/28 |  |        |  |
|        |            |             |               |         |      |               | 审核               |               |  |        |  |
| 工艺版次   | H          | 69-G0493-25 | 朱永江2025-02-28 |         |      |               | 审定               |               |  |        |  |
| A01    | 标记         | 更改单号        | 签 名           | 标记      | 更改单号 | 签 名           | 批准               | 王建朋 2025/2/28 |  |        |  |

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| 军品机加分厂 |            |          |          |         |      | 共 5 页         |      | 第 3 页    |      |        |  |
| 产品型号   |            | J/RT-148 |          | 零组件号    |      | J/RT148-02-02 |      | 零组件名称    |      | 1区计量活门 |  |
| 材料牌号   |            | 9Cr18    |          | 材料标准    |      | GJB 2294-95   |      | 毛坯类型     |      | 棒料     |  |
| 工<br>序 | 工 序<br>名 称 | 设 备      |          | 工 艺 装 备 |      |               |      | 备注       |      |        |  |
|        |            | 名称       | 型号       | 名称      | 夹具图号 | 刀具图号          | 量具图号 |          |      |        |  |
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| 35     | 清洗         | 清洗槽      |          |         |      |               |      |          |      |        |  |
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| 40     | 检验         |          |          |         |      |               |      |          |      |        |  |
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| 45     | 淬火         |          |          |         |      |               |      |          | 特殊过程 |        |  |
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| 50     | 磨外圆        | 数控无心磨    | KRONOS S |         |      |               |      |          |      |        |  |
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|        |            |          |        |         |      |               |      |          |  |        |  |
|--------|------------|----------|--------|---------|------|---------------|------|----------|--|--------|--|
| 一四三厂   |            | 综 合 目 录  |        |         |      | 编 号           |      | 69Y-3076 |  |        |  |
| 军品机加分厂 |            |          |        |         |      | 共 5 页         |      | 第 4 页    |  |        |  |
| 产品型号   |            | J/RT-148 |        | 零组件号    |      | J/RT148-02-02 |      | 零组件名称    |  | 1区计量活门 |  |
| 材料牌号   |            | 9Cr18    |        | 材料标准    |      | GJB 2294-95   |      | 毛坯类型     |  | 棒料     |  |
| 工<br>序 | 工 序<br>名 称 | 设 备      |        | 工 艺 装 备 |      |               |      | 备注       |  |        |  |
|        |            | 名称       | 型号     | 名称      | 夹具图号 | 刀具图号          | 量具图号 |          |  |        |  |
|        |            | 无心磨      | M1080B |         |      |               |      |          |  |        |  |
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| 55     | 去毛刺        | 钳工台      |        |         |      |               |      |          |  |        |  |
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| 60     | 清洗         | 清洗槽      |        |         |      |               |      |          |  |        |  |
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| 65     | 磁粉检查       |          |        |         |      |               |      |          |  |        |  |
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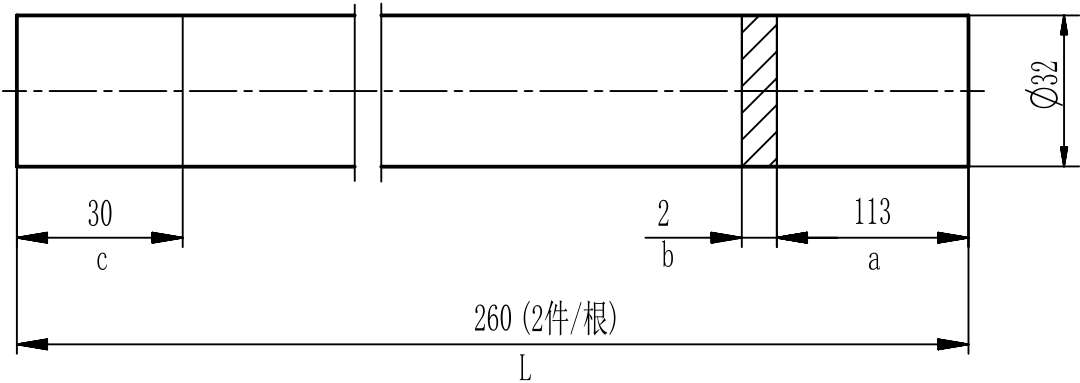
工艺

PDM 系统有效文件

|        |            |          |    |         |      |               |      |          |  |        |  |
|--------|------------|----------|----|---------|------|---------------|------|----------|--|--------|--|
| 一四三厂   |            | 综 合 目 录  |    |         |      | 编 号           |      | 69Y-3076 |  |        |  |
| 军品机加分厂 |            |          |    |         |      | 共 5 页         |      | 第 5 页    |  |        |  |
| 产品型号   |            | J/RT-148 |    | 零组件号    |      | J/RT148-02-02 |      | 零组件名称    |  | 1区计量活门 |  |
| 材料牌号   |            | 9Cr18    |    | 材料标准    |      | GJB 2294-95   |      | 毛坯类型     |  | 棒料     |  |
| 工<br>序 | 工 序<br>名 称 | 设 备      |    | 工 艺 装 备 |      |               |      | 备注       |  |        |  |
|        |            | 名称       | 型号 | 名称      | 夹具图号 | 刀具图号          | 量具图号 |          |  |        |  |
|        |            |          |    |         |      |               |      |          |  |        |  |
|        |            |          |    |         |      |               |      |          |  |        |  |
| 70     | 烧伤检查       |          |    |         |      |               |      |          |  |        |  |
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| 75     | 清洗         | 清洗槽      |    |         |      |               |      |          |  |        |  |
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| 80     | 检验         |          |    |         |      |               |      |          |  |        |  |
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| 85     | 包装         |          |    |         |      |               |      |          |  |        |  |
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PDM 系统有效文件

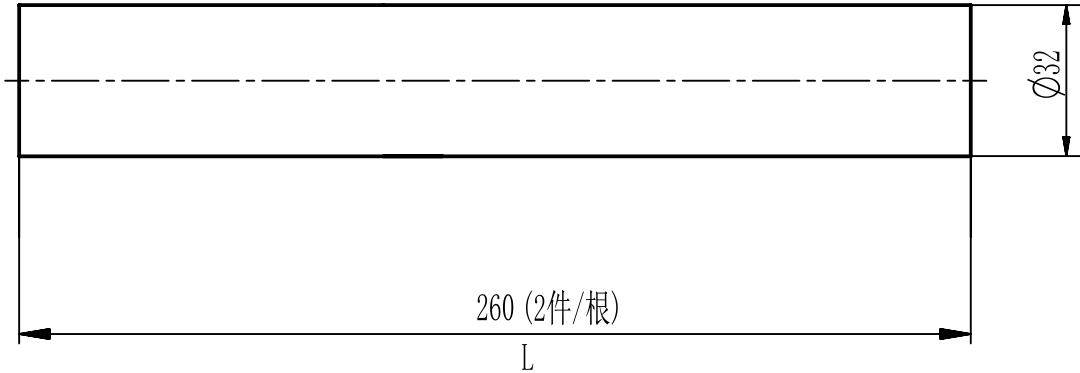
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|------------|----------|---------|---------------|--|-------|--------|------|----------|-------|
| 一四三厂       |          | 工 序 卡 片 |               |  |       |        | 编 号  | 69Y-3076 |       |
| 军品机加分厂     |          |         |               |  |       |        | 工序号  | 0        |       |
| 产品型号       | J/RT-148 | 零组件号    | J/RT148-02-02 |  | 零组件名称 | 1区计量活门 |      | 共 1 页    | 第 1 页 |
| 毛坯类型       | 棒料       | 硬度      |               |  | 设备名称  | 设备型号   | 程序编号 |          |       |
| 材料牌号       | 9Cr18    |         |               |  |       |        |      |          |       |
| 工 序<br>名 称 | 下料       |         |               |  |       |        |      |          |       |
|            |          |         |               |  |       |        |      |          |       |



| 工步        | 加 工 内 容                 | 夹具图号 | 刀具图号 | 量 具 |      | 技 术 要 求                    |    |               |
|-----------|-------------------------|------|------|-----|------|----------------------------|----|---------------|
|           |                         |      |      | 序 号 | 量具图号 |                            |    |               |
| 1         | 按图下料: Φ32×260<br>(2件/根) |      |      |     |      | 1. 材料标准:<br>GJB 2294-95    |    |               |
|           |                         |      |      |     |      | 2. 允许按单件计算下<br>料尺寸和发料      |    |               |
|           |                         |      |      |     |      | 单件(n)下料尺寸:<br>$L=n(a+b)+c$ |    |               |
|           |                         |      |      |     |      | 3. 带Φ32X80试样件<br>1根。       |    |               |
| 会 签       |                         |      |      |     |      |                            |    |               |
| 王国像       |                         |      |      |     | 编制   | 高景山 2023-9-11              |    |               |
| 2023-9-12 |                         |      |      |     | 校对   | 许洪洋 2023-9-11              |    |               |
|           |                         |      |      |     | 标审   | 刘阳松 2023-9-12              |    |               |
|           |                         |      |      |     | 审核   |                            |    |               |
| 工艺版次      |                         |      |      |     | 审定   |                            |    |               |
| A         | 标记                      | 更改单号 | 签 名  | 标记  | 更改单号 | 签 名                        | 批准 | 王建朋 2023-9-12 |

工艺  
PDM 系统有效文件

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|------------|----------|---------|---------------|-------|--------|------|----------|----------|
| 一四三厂       |          | 协 作 卡 片 |               |       |        | 编 号  | 69Y-3076 |          |
| 军品机加分厂     |          |         |               |       |        | 工序号  | 5        |          |
| 产品型号       | J/RT-148 | 零组件号    | J/RT148-02-02 | 零组件名称 | 1区计量活门 |      | 共 1 页    | 第 1 页    |
| 工 序<br>名 称 | 调质       |         |               | 毛坯类型  | 棒料     | 协作单位 | 军品热表分厂   |          |
|            |          |         |               | 材料牌号  | 9Cr18  |      | 硬度       | 27~35HRC |



| 序号 | 技 术 要 求                | 备 注                             |
|----|------------------------|---------------------------------|
|    | 调质<br><br>硬度为：27~35HRC | 1. 带1件 $\Phi 32 \times 80$ 试样件。 |

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| 初样研制      |
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|    |      |     |    |      |     |    | 编制            | 高景山 | 2023-9-11 |
|    |      |     |    |      |     |    | 校对            | 许洪洋 | 2023-9-11 |
|    |      |     |    |      |     |    | 标审            | 刘阳松 | 2023-9-12 |
|    |      |     |    |      |     |    | 审核            |     |           |
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## PDM 系统有效文件

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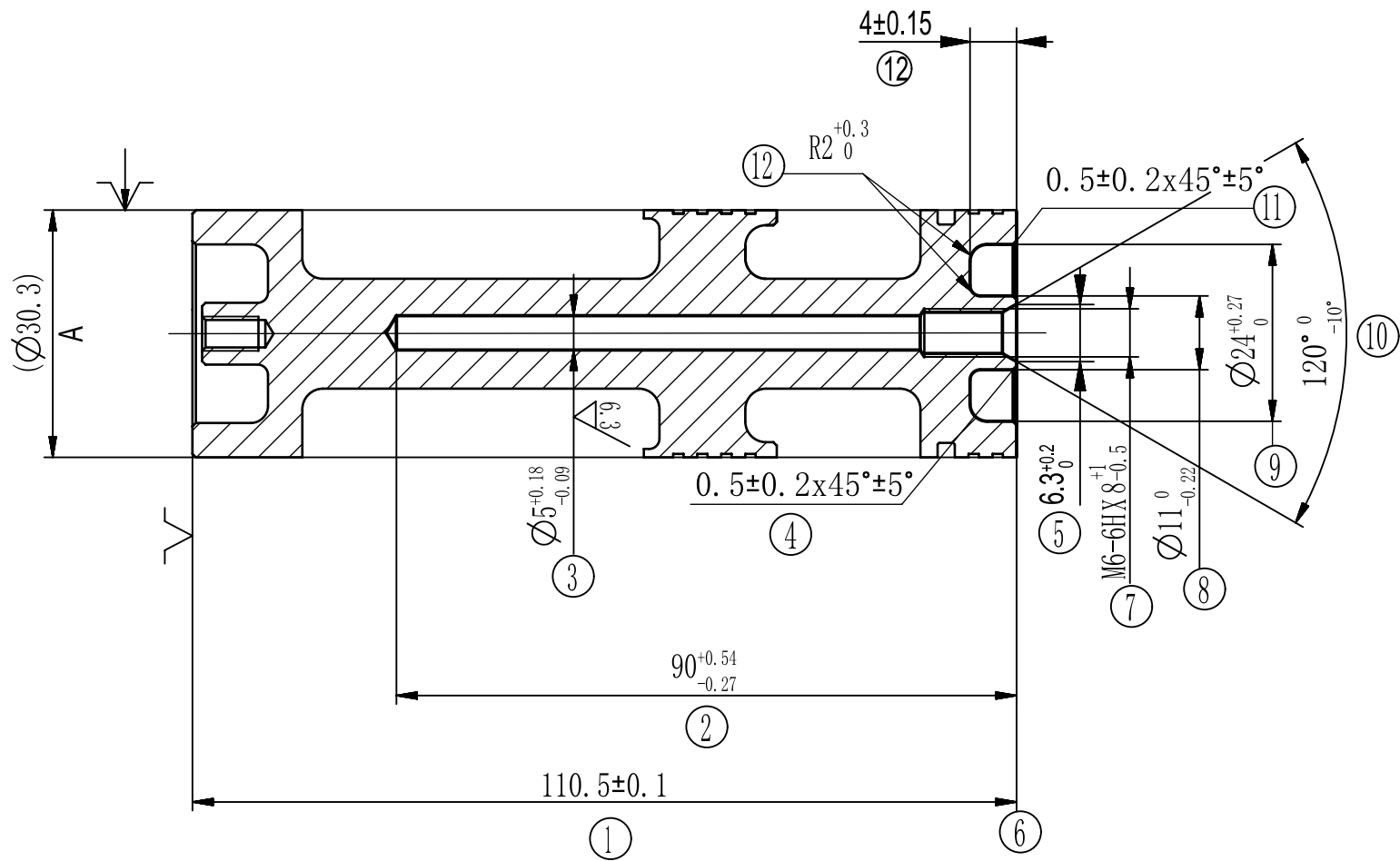


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工序卡片

编号 69Y-3076  
工序号 15

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|------|----------|------|---------------|-------|---------------|----------------------|-------|
| 产品型号 | J/RT-148 | 零组件号 | J/RT148-02-02 | 零组件名称 | 1区计量活门        | 共 1 页                | 第 1 页 |
| 毛坯类型 | 棒料       | 硬度   | 27~35HRC      | 设备名称  | 设备型号          | 程序编号                 |       |
| 材料牌号 | 9Cr18    |      |               | 数控车床  | CTX310V3      | J/RT148-02-02-GX15-1 |       |
| 工序名称 | 车零件      |      |               | 数控车床  | GS 200/66PLUS | J/RT148-02-02-GX15-2 |       |
|      |          |      |               | 数控车床  | CTX310V1      | J/RT148-02-02-GX15-3 |       |

| 工步 | 加工内容                       | 夹具图号 | 刀具图号 | 量具  |                 |
|----|----------------------------|------|------|-----|-----------------|
|    |                            |      |      | 序号  | 量具图号            |
| 1  | 平端面⑥                       | 软爪   |      | ①②⑤ | 卡尺0.02          |
|    |                            |      |      | ③   | 0220/5HJ12 T,Z  |
|    |                            |      |      | ④⑪  | 轮廓仪             |
| 2  | 车槽⑨                        |      |      | ⑦   | 0421/M6-6H T,Z  |
|    |                            |      |      |     | 0228/M6-H-II T  |
|    |                            |      |      |     | 0228/M6-6H-II Z |
| 3  | 倒圆⑫                        |      |      |     | 卡尺0.02          |
|    |                            |      |      | ⑧⑨  | 卡尺0.02          |
|    |                            |      |      | ⑩   | 轮廓仪             |
| 4  | 钻孔③                        |      |      | ⑫   | 轮廓仪             |
|    | 螺纹底孔 $\phi 5.08^{+0.15}_0$ |      |      | 粗糙度 | 粗糙度对比样块         |
| 5  | 攻螺纹⑦                       |      |      |     |                 |
| 6  | 锐边磨圆R0.1~R0.3              |      |      |     |                 |

- 技术要求
- 外圆A保持光洁锐边
  - 尺寸⑫首件解剖计量合格后由合格程序保证。

|    |     |           |
|----|-----|-----------|
| 编制 | 朱永江 | 2025-2-28 |
| 校对 | 胡文杰 | 2025-2-28 |
| 标审 | 袁贻乾 | 2025-2-28 |
| 审核 |     |           |
| 审定 |     |           |

|    |      |     |    |      |     |    |     |           |
|----|------|-----|----|------|-----|----|-----|-----------|
| 标记 | 更改单号 | 签 名 | 标记 | 更改单号 | 签 名 | 批准 | 王建朋 | 2025-2-28 |
|----|------|-----|----|------|-----|----|-----|-----------|

初样研制

会 签

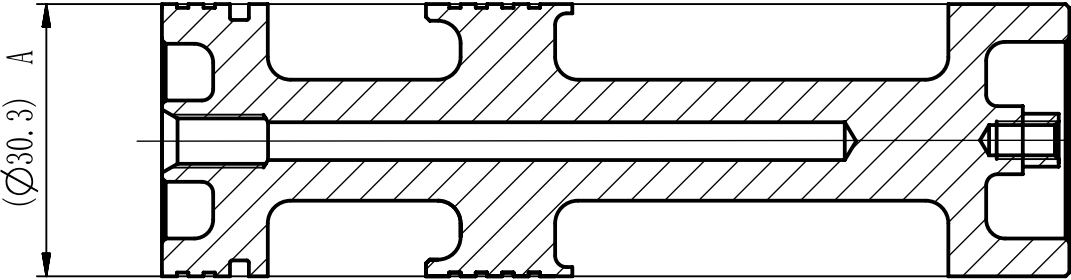
工艺版次

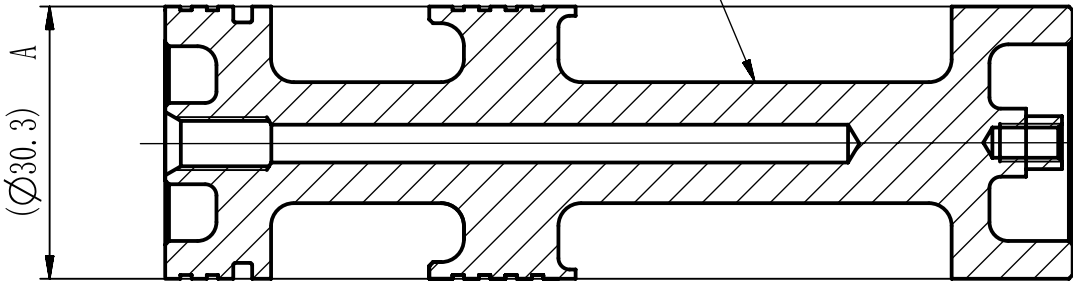
A01

其余  $\frac{3.2}{\nabla}$

|            |  |          |  |      |  |               |  |        |  |          |  |                      |  |       |  |
|------------|--|----------|--|------|--|---------------|--|--------|--|----------|--|----------------------|--|-------|--|
| 一 四 三 厂    |  | 工 序 卡 片  |  |      |  |               |  | 编 号    |  | 69Y-3076 |  |                      |  |       |  |
| 军品机加分厂     |  |          |  |      |  |               |  | 工 序 号  |  | 20       |  |                      |  |       |  |
| 产品型号       |  | J/RT-148 |  | 零组件号 |  | J/RT148-02-02 |  | 零组件名称  |  | 1区计量活门   |  | 共 1 页                |  | 第 1 页 |  |
| 毛坯类型       |  | 棒料       |  | 硬度   |  | 27~35HRC      |  | 设备名称   |  | 设备型号     |  | 程序编号                 |  |       |  |
| 材料牌号       |  | 9Cr18    |  |      |  |               |  | 小型钻孔中心 |  | TC—S2Z   |  | J/RT148-02-02-GX20-1 |  |       |  |
| 工 序<br>名 称 |  | 铣槽       |  |      |  |               |  |        |  |          |  |                      |  |       |  |
|            |  |          |  |      |  |               |  |        |  |          |  |                      |  |       |  |
| 工步         |  | 加 工 内 容  |  |      |  | 夹具图号          |  | 刀具图号   |  | 量 具      |  |                      |  |       |  |
|            |  |          |  |      |  |               |  |        |  | 序 号      |  | 量具图号                 |  |       |  |
| 1          |  | 铣槽 ②     |  |      |  |               |  |        |  | ① ②      |  | 卡尺0.02               |  |       |  |
|            |  |          |  |      |  |               |  |        |  | 粗糙度      |  | 粗糙度对比样块              |  |       |  |
|            |  |          |  |      |  |               |  |        |  |          |  |                      |  |       |  |
|            |  |          |  |      |  |               |  |        |  |          |  |                      |  |       |  |
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|            |  |          |  |      |  |               |  |        |  |          |  |                      |  |       |  |
|            |  |          |  |      |  |               |  |        |  |          |  |                      |  |       |  |
|            |  |          |  |      |  |               |  |        |  |          |  |                      |  |       |  |
|            |  |          |  |      |  |               |  |        |  |          |  |                      |  |       |  |
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|            |  |          |  |      |  |               |  |        |  |          |  |                      |  |       |  |
|            |  |          |  |      |  |               |  |        |  |          |  |                      |  |       |  |
|            |  |          |  |      |  |               |  |        |  |          |  |                      |  |       |  |

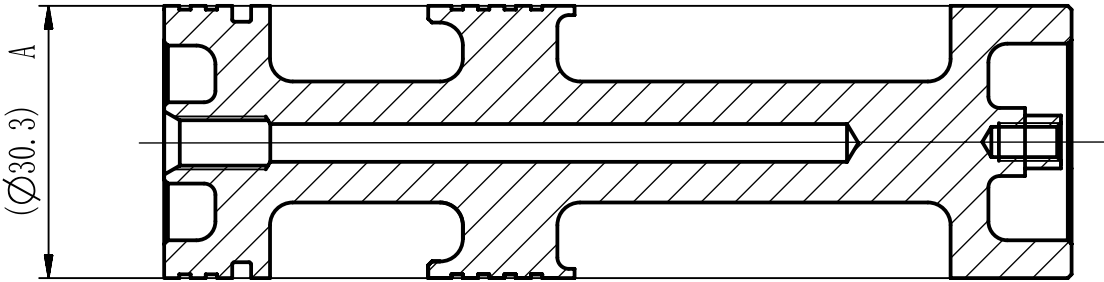
工艺  
PDM 系统有效文件

|                                                                                               |                |         |          |               |         |                   |        |               |       |       |
|-----------------------------------------------------------------------------------------------|----------------|---------|----------|---------------|---------|-------------------|--------|---------------|-------|-------|
| 一四三厂                                                                                          |                | 工 序 卡 片 |          |               |         |                   | 编 号    | 69Y-3076      |       |       |
| 军品机加分厂                                                                                        |                |         |          |               |         |                   | 工 序 号  | 25            |       |       |
| 产 品 型 号                                                                                       | J/RT-148       |         | 零 组 件 号  | J/RT148-02-02 |         | 零 组 件 名 称         | 1区计量活门 |               | 共 1 页 | 第 1 页 |
| 毛 坯 类 型                                                                                       | 棒料             | 硬 度     | 27~35HRC |               | 设备名称    | 设备型号              | 程序编号   |               |       |       |
| 材 料 牌 号                                                                                       | 9Cr18          |         |          |               | 钳工台     |                   |        |               |       |       |
| 工 序<br>名 称                                                                                    | 去毛刺            |         |          |               |         |                   |        |               |       |       |
|                                                                                               |                |         |          |               |         |                   |        |               |       |       |
| <div></div> |                |         |          |               |         |                   |        |               |       |       |
| 工 步                                                                                           | 加 工 内 容        | 夹 具 图 号 | 刀 具 图 号  | 量 具           |         | 技 术 要 求           |        |               |       |       |
|                                                                                               |                |         |          | 序 号           | 量 具 图 号 |                   |        |               |       |       |
| 1                                                                                             | 按〈69-GZ7〉去毛刺操作 |         |          |               |         | 1. 外圆A保持光洁<br>锐边。 |        |               |       |       |
|                                                                                               | 指导5.1.3去除内孔毛刺  |         |          |               |         |                   |        |               |       |       |
|                                                                                               |                |         |          |               |         |                   |        |               |       |       |
|                                                                                               |                |         |          |               |         |                   |        |               |       |       |
|                                                                                               |                |         |          |               |         |                   |        |               |       |       |
|                                                                                               |                |         |          |               |         |                   |        |               |       |       |
| 初样研制                                                                                          |                |         |          |               |         |                   |        |               |       |       |
|                                                                                               |                |         |          |               |         |                   |        |               |       |       |
| 会 签                                                                                           |                |         |          |               |         |                   |        |               |       |       |
|                                                                                               |                |         |          |               | 编制      | 段长超 2023-9-25     |        |               |       |       |
|                                                                                               |                |         |          |               | 校对      | 杨一峰 2023-9-26     |        |               |       |       |
|                                                                                               |                |         |          |               | 标审      | 杨科 2023-9-27      |        |               |       |       |
|                                                                                               |                |         |          |               | 审核      |                   |        |               |       |       |
| 工艺版次                                                                                          |                |         |          |               | 审定      |                   |        |               |       |       |
| A                                                                                             | 标记             | 更改单号    | 签 名      | 标记            | 更改单号    | 签 名               | 批准     | 陈云维 2023-10-9 |       |       |

|         |                                                                                                   |           |         |          |               |         |           |               |                |              |  |
|---------|---------------------------------------------------------------------------------------------------|-----------|---------|----------|---------------|---------|-----------|---------------|----------------|--------------|--|
| 工艺<br>工 | 一四三厂                                                                                              |           | 工 序 卡 片 |          |               |         |           | 编 号           | 69Y-3076       |              |  |
|         | 军品机加分厂                                                                                            |           |         |          |               |         |           | 工 序 号         | 30             |              |  |
|         | 产品型号                                                                                              | J/RT-148  |         | 零组件号     | J/RT148-02-02 |         | 零组件名称     | 1区计量活门        | 共 1 页          | 第 1 页        |  |
|         | 毛坯类型                                                                                              | 棒料        | 硬度      | 27~35HRC |               | 设备名称    | 设备型号      | 程序编号          |                |              |  |
|         | 材料牌号                                                                                              | 9Cr18     |         |          |               | 激光打标机   | YLP-HM30D |               |                |              |  |
|         | 工 序 名 称                                                                                           | 标 印       |         |          |               |         |           |               |                |              |  |
|         | <div>标印处</div>  |           |         |          |               |         |           |               |                |              |  |
|         | 工 步                                                                                               | 加 工 内 容   |         | 夹 具 图 号  |               | 刀 具 图 号 |           | 量 具           |                | 技 术 要 求      |  |
|         |                                                                                                   |           |         |          |               |         |           | 序 号           | 量具图号           |              |  |
|         | 1                                                                                                 | 在标印处标刻零件号 |         |          |               |         |           |               |                | 1. 外圆A保持光洁锐边 |  |
|         |                                                                                                   |           |         |          |               |         |           |               | 2. 采用直体字，字高1.5 |              |  |
| 初样研制    |                                                                                                   |           |         |          |               |         |           |               |                |              |  |
| 会 签     |                                                                                                   |           |         |          |               |         |           |               |                |              |  |
|         |                                                                                                   |           |         |          |               |         | 编制        | 段长超 2023-9-25 |                |              |  |
|         |                                                                                                   |           |         |          |               |         | 校对        | 杨一峰 2023-9-26 |                |              |  |
|         |                                                                                                   |           |         |          |               |         | 标审        | 杨科 2023-9-27  |                |              |  |
|         |                                                                                                   |           |         |          |               |         | 审核        |               |                |              |  |
| 工艺版次    |                                                                                                   |           |         |          |               |         | 审定        |               |                |              |  |
| A       | 标记                                                                                                | 更改单号      | 签 名     | 标记       | 更改单号          | 签 名     | 批准        | 陈云维 2023-10-9 |                |              |  |

工艺  
PDM 系统有效文件

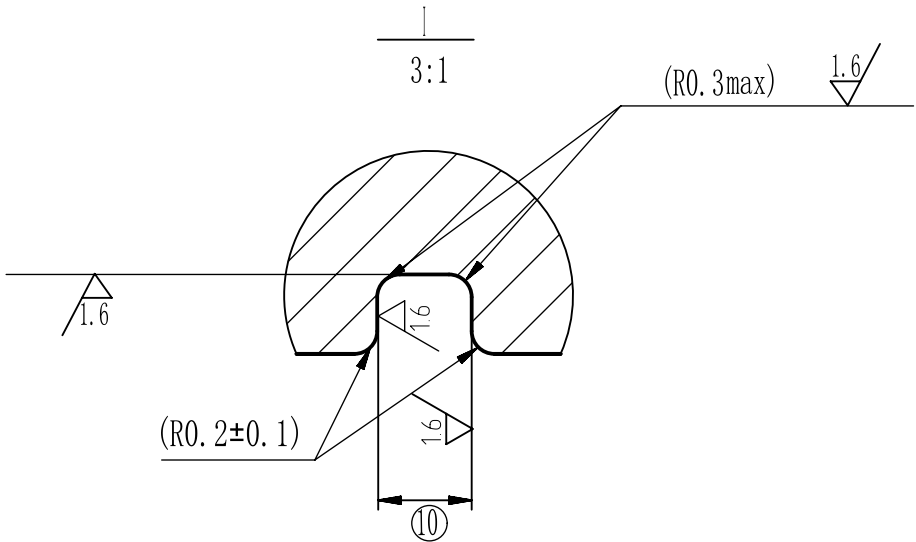
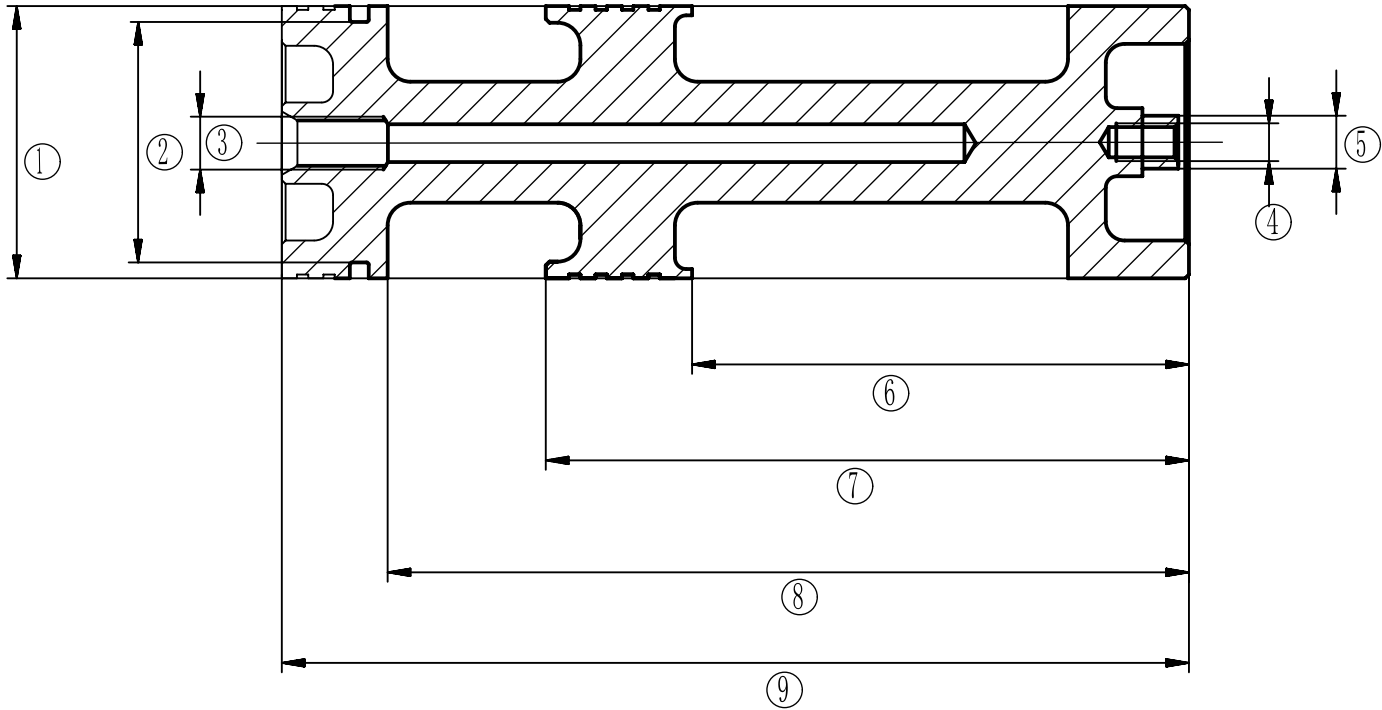
|            |          |         |          |               |         |           |         |          |       |       |
|------------|----------|---------|----------|---------------|---------|-----------|---------|----------|-------|-------|
| 一四三厂       |          | 工 序 卡 片 |          |               |         |           | 编 号     | 69Y-3076 |       |       |
| 军品机加分厂     |          |         |          |               |         |           | 工 序 号   | 35       |       |       |
| 产 品 型 号    | J/RT-148 |         | 零 组 件 号  | J/RT148-02-02 |         | 零 组 件 名 称 | 1区计量活门  |          | 共 1 页 | 第 1 页 |
| 毛 坯 类 型    | 棒 料      | 硬 度     | 27~35HRC |               | 设 备 名 称 | 设 备 型 号   | 程 序 编 号 |          |       |       |
| 材 料 牌 号    | 9Cr18    |         |          |               | 清 洗 槽   |           |         |          |       |       |
| 工 序<br>名 称 | 清 洗      |         |          |               |         |           |         |          |       |       |
|            |          |         |          |               |         |           |         |          |       |       |



| 工 步  | 加 工 内 容                                | 夹 具 图 号 | 刀 具 图 号 | 量 具 |         | 技 术 要 求       |    |     |           |
|------|----------------------------------------|---------|---------|-----|---------|---------------|----|-----|-----------|
|      |                                        |         |         | 序 号 | 量 具 图 号 |               |    |     |           |
| 1    | 用毛刷在溶剂清洗剂中将零件内外表面的附着物涮洗干净，并放入干净的零件专用盒。 |         |         |     |         | 1. 外圆A保持光洁锐边。 |    |     |           |
|      |                                        |         |         |     |         |               |    |     |           |
|      |                                        |         |         |     |         |               |    |     |           |
|      |                                        |         |         |     |         |               |    |     |           |
|      |                                        |         |         |     |         |               |    |     |           |
| 初样研制 |                                        |         |         |     |         |               |    |     |           |
|      |                                        |         |         |     |         |               |    |     |           |
| 会 签  |                                        |         |         |     |         |               |    |     |           |
|      |                                        |         |         |     |         |               |    |     |           |
|      |                                        |         |         |     |         |               |    |     |           |
|      |                                        |         |         |     |         |               |    |     |           |
|      |                                        |         |         |     |         |               |    |     |           |
| 工艺版次 |                                        |         |         |     |         |               |    |     |           |
| A    | 标记                                     | 更改单号    | 签 名     | 标记  | 更改单号    | 签 名           | 编制 | 段长超 | 2023-9-25 |
|      |                                        |         |         |     |         |               | 校对 | 杨一峰 | 2023-9-26 |
|      |                                        |         |         |     |         |               | 标审 | 杨科  | 2023-9-27 |
|      |                                        |         |         |     |         |               | 审核 |     |           |
|      |                                        |         |         |     |         |               | 审定 |     |           |
|      |                                        |         |         |     |         |               | 批准 | 陈云维 | 2023-10-9 |

工艺  
PDM系统有效文件

其余  $\sqrt{3.2}$



|         |                                            |             |                 |      |       |         |               |          |       |
|---------|--------------------------------------------|-------------|-----------------|------|-------|---------|---------------|----------|-------|
| 一 四 三 厂 |                                            | 检 验 工 序 卡 片 |                 |      |       |         | 编 号           | 69Y-3076 |       |
| 军品机加分厂  |                                            |             |                 |      |       |         | 工 序 号         | 40       |       |
| 产品型号    | J/RT-148                                   | 零组件号        | J/RT148-02-02   |      | 零组件名称 | 1区计量活门  |               | 共 1 页    | 第 1 页 |
| 材料牌号    | 9Cr18                                      |             |                 | 毛坯类型 | 棒料    | 硬 度     |               | 27~35HRC |       |
| 编号      | 检 验 内 容                                    |             | 量 具 图 号         |      | 编号    | 检 验 内 容 |               | 量 具 图 号  |       |
| I       | 外观检查                                       |             |                 |      |       |         |               |          |       |
| 1       | 检查零件应无碰伤, 压                                |             |                 |      |       |         |               |          |       |
|         | 伤, 划伤, 锈蚀等缺陷                               |             |                 |      |       |         |               |          |       |
| 2       | 检查零件表面粗糙度                                  |             | 糙度比较样块          |      |       |         |               |          |       |
| 3       | 用10倍放大镜检查零                                 |             |                 |      |       |         |               |          |       |
|         | 件表面应无毛刺                                    |             |                 |      |       |         |               |          |       |
|         |                                            |             |                 |      |       |         |               |          |       |
| II      | 尺寸检查                                       |             |                 |      |       |         |               |          |       |
| ①       | Ø30.3 <sup>0</sup> <sub>-0.05</sub>        |             | 千分尺0.01         |      |       |         |               |          |       |
| ②       | Ø25.8 <sup>0</sup> <sub>-0.052</sub>       |             | 刀口千分尺0.01       |      |       |         |               |          |       |
| ③       | M6-6H                                      |             | 0420/M6-6H T, Z |      |       |         |               |          |       |
| ④       | M4-6H                                      |             | 0420/M4-6H T, Z |      |       |         |               |          |       |
| ⑤       | 8 <sup>0</sup> <sub>-0.22</sub>            |             | 投影              |      |       |         |               |          |       |
| ⑥       | 64±0.05                                    |             | 投影              |      |       |         |               |          |       |
| ⑦       | 76.5±0.1                                   |             | 卡尺0.02          |      |       |         |               |          |       |
| ⑧       | 100.5±0.1                                  |             | 投影              |      |       |         |               |          |       |
| ⑨       | 110.5±0.1                                  |             | 卡尺0.02          |      |       |         |               |          |       |
| ⑩       | 2.5 <sup>+0.1</sup> <sub>0</sub>           |             | 0340/2H12 TZ    |      |       |         |               |          |       |
|         |                                            |             |                 |      |       |         |               |          |       |
|         |                                            |             |                 |      |       |         |               |          |       |
|         |                                            |             |                 |      |       |         |               |          |       |
| 技 术 要 求 | 1. 外圆①保持光洁锐边<br>2. 尺寸①②⑥⑦⑧⑨⑩ 允许在光学轴类测量仪上检查 |             |                 |      |       |         |               |          |       |
|         |                                            |             |                 |      |       | 编制      | 段长超 2023-9-25 |          |       |
|         |                                            |             |                 |      |       | 校对      | 杨一峰 2023-9-26 |          |       |
|         |                                            |             |                 |      |       | 标审      | 杨科 2023-9-27  |          |       |
|         |                                            |             |                 |      |       | 审核      |               |          |       |
|         |                                            |             |                 |      |       | 审定      |               |          |       |
| 标记      | 更改单号                                       | 签 名         | 标记              | 更改单号 | 签 名   | 批准      | 陈云维 2023-10-9 |          |       |

初样研制

会 签

吴昊

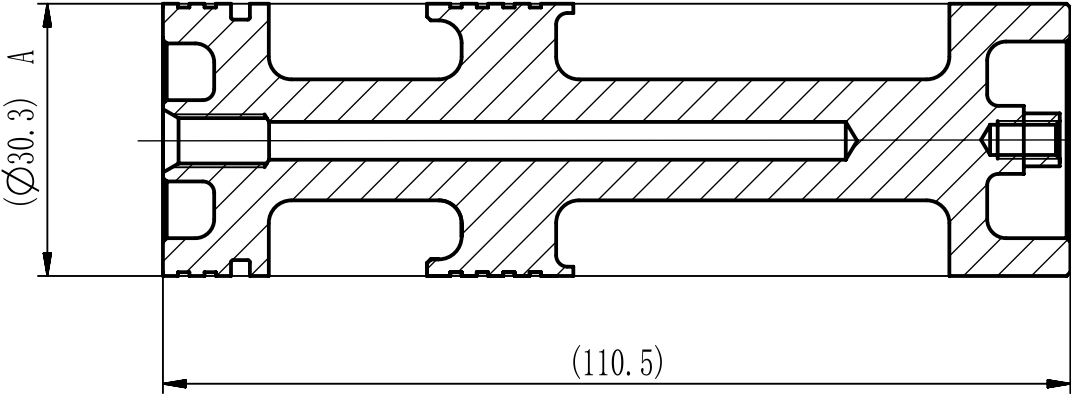
2023-9-27

工艺版次

A

工艺  
PDM 系统有效文件

|            |          |            |               |       |        |      |          |        |
|------------|----------|------------|---------------|-------|--------|------|----------|--------|
| 一四三厂       |          | 协  作  卡  片 |               |       |        | 编  号 | 69Y-3076 |        |
| 军品机加分厂     |          |            |               |       |        | 工序号  | 45       |        |
| 产品型号       | J/RT-148 | 零组件号       | J/RT148-02-02 | 零组件名称 | 1区计量活门 |      | 共 1 页    | 第 1 页  |
| 工 序<br>名 称 | 淬 火      |            |               | 毛坯类型  | 棒料     | 协作单位 | 军品热表分厂   |        |
|            |          |            |               | 材料牌号  | 9Cr18  |      | 硬度       | ≥56HRC |



| 序号 | 技 术 要 求                   | 备 注                              |
|----|---------------------------|----------------------------------|
|    | 真空淬火：<br><br>硬度要求为：≥56HRC | 1. 表面A保持光洁锐边<br>2. 带Ø10x10试样件，5件 |

初样研制

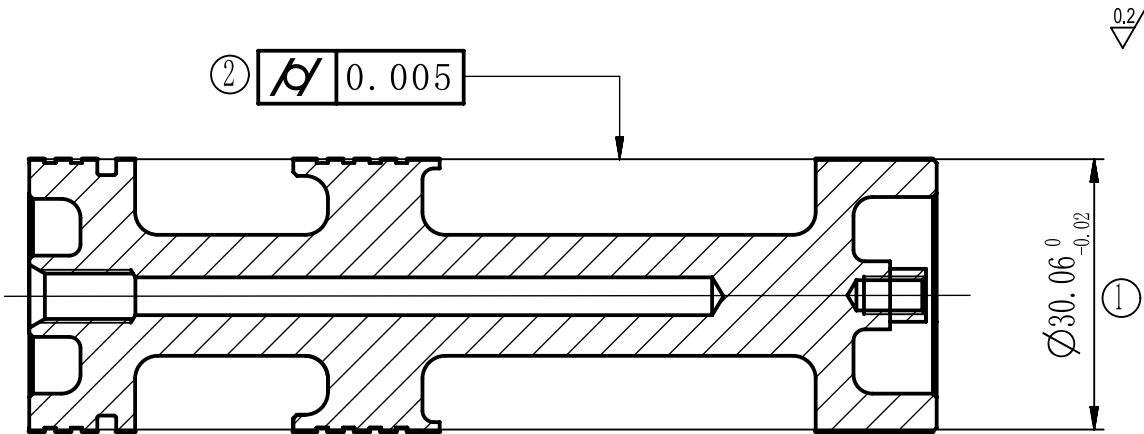
会 签

|           |    |      |      |    |      |      |     |               |
|-----------|----|------|------|----|------|------|-----|---------------|
| 陈剑先       |    |      |      |    |      | 编制   | 段长超 | 2023-9-25     |
| 2023-9-27 |    |      |      |    |      | 校对   | 杨一峰 | 2023-9-26     |
|           |    |      |      |    |      | 标审   | 杨科  | 2023-9-27     |
|           |    |      |      |    |      | 审核   |     |               |
| 工艺版次      |    |      |      |    |      | 审定   |     |               |
| A         | 标记 | 更改单号 | 签  名 | 标记 | 更改单号 | 签  名 | 批准  | 陈云维 2023-10-9 |



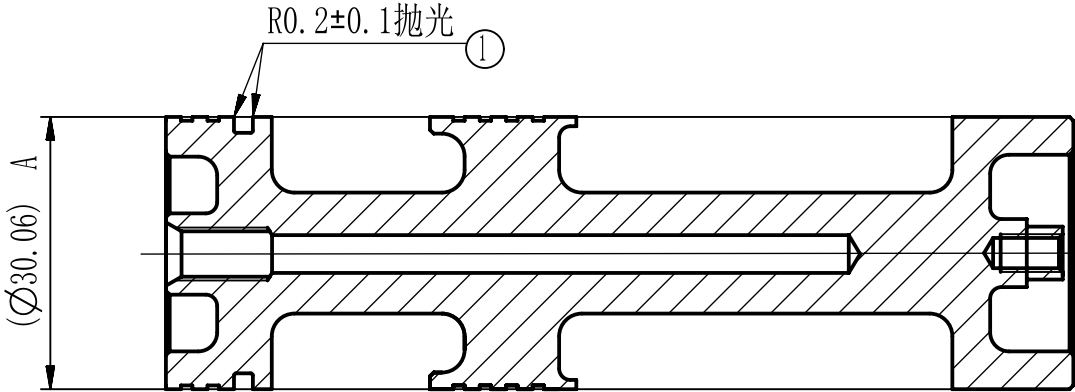
工艺  
PDM 系统有效文件

|         |          |         |         |               |  |           |          |                      |       |       |
|---------|----------|---------|---------|---------------|--|-----------|----------|----------------------|-------|-------|
| 一四三厂    |          | 工 序 卡 片 |         |               |  |           | 编 号      | 69Y-3076             |       |       |
| 军品机加分厂  |          |         |         |               |  |           | 工 序 号    | 50                   |       |       |
| 产 品 型 号 | J/RT-148 |         | 零 组 件 号 | J/RT148-02-02 |  | 零 组 件 名 称 | 1区计量活门   |                      | 共 1 页 | 第 1 页 |
| 毛 坯 类 型 | 棒料       |         | 硬 度     | ≥56HRC        |  | 设 备 名 称   | 设 备 型 号  | 程 序 编 号              |       |       |
| 材 料 牌 号 | 9Cr18    |         |         |               |  | 数 控 无 心 磨 | KRONOS S | J/RT148-02-02-GX50-1 |       |       |
| 工 序 名 称 | 磨 外 圆    |         |         |               |  | 无 心 磨     | M1080B   |                      |       |       |
|         |          |         |         |               |  |           |          |                      |       |       |

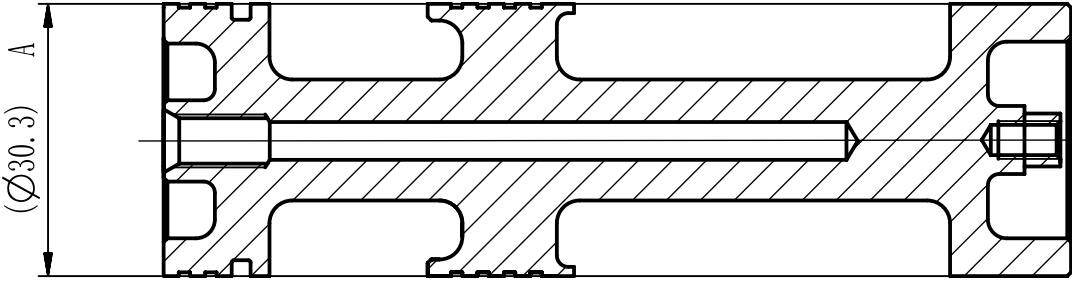


|           | 工步 | 加 工 内 容            | 夹具图号 |    | 刀具图号           |     | 量 具      |                                                         | 技 术 要 求 |
|-----------|----|--------------------|------|----|----------------|-----|----------|---------------------------------------------------------|---------|
|           |    |                    |      |    |                |     | 序 号      | 量具图号                                                    |         |
|           | 1  | 粗磨外圆 ① ， 加工至Φ30.1  |      |    | 砂轮（400X20X127） | ①②  | 块规       | 1. 外圆①保持光洁锐边。                                           |         |
|           |    |                    |      |    |                |     | 指示尺0.001 |                                                         |         |
|           |    |                    |      |    |                | 粗糙度 | 粗糙度测量仪   | 2. 加工时要求砂轮线速度为35m/s, 粗磨进给量为0.05~0.1mm, 半精磨为0.02~0.05mm, |         |
|           | 2  | 半精磨外圆 ①， 加工至Φ30.07 |      |    |                |     |          | 精磨为0.005~0.01mm。                                        |         |
|           |    |                    |      |    |                |     |          | 3. 首件检查零件表面应无烧伤和裂纹。                                     |         |
|           |    |                    |      |    |                |     |          | 按MT-986检查                                               |         |
| 初样研制      | 3  | 精磨外圆 ①             |      |    |                |     |          |                                                         |         |
| 会 签       |    |                    |      |    |                |     |          |                                                         |         |
| 王普照       |    |                    |      |    |                |     | 编制       | 段长超 2023-9-25                                           |         |
| 2023-9-27 |    |                    |      |    |                |     | 校对       | 杨一峰 2023-9-26                                           |         |
|           |    |                    |      |    |                |     | 标审       | 杨科 2023-9-27                                            |         |
|           |    |                    |      |    |                |     | 审核       |                                                         |         |
| 工艺版次      |    |                    |      |    |                |     | 审定       |                                                         |         |
| A         | 标记 | 更改单号               | 签 名  | 标记 | 更改单号           | 签 名 | 批准       | 陈云维 2023-10-9                                           |         |

工艺  
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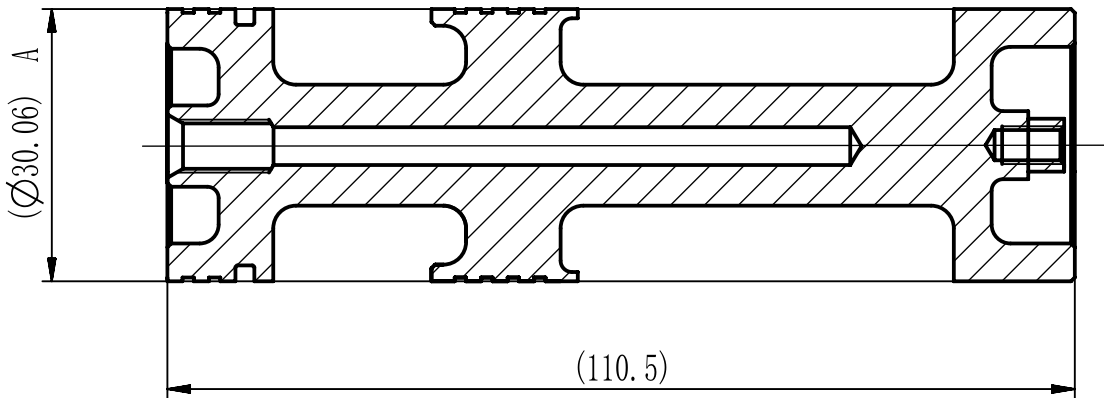
|                                                                                               |                |         |         |               |         |            |         |               |       |       |
|-----------------------------------------------------------------------------------------------|----------------|---------|---------|---------------|---------|------------|---------|---------------|-------|-------|
| 一四三厂                                                                                          |                | 工 序 卡 片 |         |               |         |            | 编 号     | 69Y-3076      |       |       |
| 军品机加分厂                                                                                        |                |         |         |               |         |            | 工 序 号   | 55            |       |       |
| 产 品 型 号                                                                                       | J/RT-148       |         | 零 组 件 号 | J/RT148-02-02 |         | 零 组 件 名 称  | 1区计量活门  |               | 共 1 页 | 第 1 页 |
| 毛 坯 类 型                                                                                       | 棒 料            | 硬 度     | ≥56HRC  |               | 设 备 名 称 | 设 备 型 号    | 程 序 编 号 |               |       |       |
| 材 料 牌 号                                                                                       | 9Cr18          |         |         |               | 钳 工 台   |            |         |               |       |       |
| 工 序<br>名 称                                                                                    | 去毛刺            |         |         |               |         |            |         |               |       |       |
|                                                                                               |                |         |         |               |         |            |         |               |       |       |
| <div></div> |                |         |         |               |         |            |         |               |       |       |
| 工 步                                                                                           | 加 工 内 容        | 夹 具 图 号 | 刀 具 图 号 | 量 具           |         | 技 术 要 求    |         |               |       |       |
|                                                                                               |                |         |         | 序 号           | 量 具 图 号 |            |         |               |       |       |
| 1                                                                                             | 按〈69-GZ7〉去毛刺操作 |         |         |               |         | 1. 外圆A保持光洁 |         |               |       |       |
|                                                                                               | 指导5.1.1去除外形毛刺  |         |         |               |         |            | 锐边。     |               |       |       |
| 2                                                                                             | 抛光箭示处          |         |         |               |         |            |         |               |       |       |
|                                                                                               |                |         |         |               |         |            |         |               |       |       |
|                                                                                               |                |         |         |               |         |            |         |               |       |       |
| 初样研制                                                                                          |                |         |         |               |         |            |         |               |       |       |
|                                                                                               |                |         |         |               |         |            |         |               |       |       |
| 会 签                                                                                           |                |         |         |               |         |            |         |               |       |       |
|                                                                                               |                |         |         |               |         | 编制         | 段长超     | 2023-9-25     |       |       |
|                                                                                               |                |         |         |               |         | 校对         | 杨一峰     | 2023-9-26     |       |       |
|                                                                                               |                |         |         |               |         | 标审         | 杨科      | 2023-9-27     |       |       |
|                                                                                               |                |         |         |               |         | 审核         |         |               |       |       |
| 工艺版次                                                                                          |                |         |         |               |         | 审定         |         |               |       |       |
| A                                                                                             | 标记             | 更改单号    | 签 名     | 标记            | 更改单号    | 签 名        | 批准      | 陈云维 2023-10-9 |       |       |

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|                                                                                               |               |         |         |               |         |                   |               |               |       |       |
|-----------------------------------------------------------------------------------------------|---------------|---------|---------|---------------|---------|-------------------|---------------|---------------|-------|-------|
| 一四三厂                                                                                          |               | 工 序 卡 片 |         |               |         |                   | 编 号           | 69Y-3076      |       |       |
| 军品机加分厂                                                                                        |               |         |         |               |         |                   | 工序号           | 60            |       |       |
| 产品型号                                                                                          | J/RT-148      |         | 零组件号    | J/RT148-02-02 |         | 零组件名称             | 1区计量活门        |               | 共 1 页 | 第 1 页 |
| 毛坯类型                                                                                          | 棒料            | 硬度      | ≥56HRC  |               | 设备名称    | 设备型号              | 程序编号          |               |       |       |
| 材料牌号                                                                                          | 9Cr18         |         |         |               | 清洗槽     |                   |               |               |       |       |
| 工 序<br>名 称                                                                                    | 清 洗           |         |         |               |         |                   |               |               |       |       |
|                                                                                               |               |         |         |               |         |                   |               |               |       |       |
| <div></div> |               |         |         |               |         |                   |               |               |       |       |
| 工 步                                                                                           | 加 工 内 容       | 夹 具 图 号 | 刀 具 图 号 | 量 具           |         | 技 术 要 求           |               |               |       |       |
|                                                                                               |               |         |         | 序 号           | 量 具 图 号 |                   |               |               |       |       |
| 1                                                                                             | 用毛刷在溶剂清洗剂中将零件 |         |         |               |         | 1. 外圆A保持光洁<br>锐边。 |               |               |       |       |
|                                                                                               | 内外表面的附着物涮洗干   |         |         |               |         |                   |               |               |       |       |
|                                                                                               | 净, 并放入干净的零件专  |         |         |               |         |                   |               |               |       |       |
|                                                                                               | 用盒。           |         |         |               |         |                   |               |               |       |       |
|                                                                                               |               |         |         |               |         |                   |               |               |       |       |
|                                                                                               |               |         |         |               |         |                   |               |               |       |       |
| 初样研制                                                                                          |               |         |         |               |         |                   |               |               |       |       |
|                                                                                               |               |         |         |               |         |                   |               |               |       |       |
| 会 签                                                                                           |               |         |         |               |         |                   |               |               |       |       |
|                                                                                               |               |         |         |               |         | 编制                | 段长超 2023-9-25 |               |       |       |
|                                                                                               |               |         |         |               |         | 校对                | 杨一峰 2023-9-26 |               |       |       |
|                                                                                               |               |         |         |               |         | 标审                | 杨科 2023-9-27  |               |       |       |
|                                                                                               |               |         |         |               |         | 审核                |               |               |       |       |
| 工艺版次                                                                                          |               |         |         |               |         | 审定                |               |               |       |       |
| A                                                                                             | 标记            | 更改单号    | 签 名     | 标记            | 更改单号    | 签 名               | 批准            | 陈云维 2023-10-9 |       |       |

工艺  
PDM 系统有效文件

|            |          |         |               |           |        |         |          |        |
|------------|----------|---------|---------------|-----------|--------|---------|----------|--------|
| 一 四 三 厂    |          | 协 作 卡 片 |               |           |        | 编 号     | 69Y-3076 |        |
| 军品机加分厂     |          |         |               |           |        | 工序号     | 65       |        |
| 产 品 型 号    | J/RT-148 | 零 组 件 号 | J/RT148-02-02 | 零 组 件 名 称 | 1区计量活门 |         | 共 1 页    | 第 1 页  |
| 工 序<br>名 称 | 磁粉检查     |         |               | 毛 坯 类 型   | 棒 料    | 协 作 单 位 | 质 量 部    |        |
|            |          |         |               | 材 料 牌 号   | 9Cr18  |         | 硬 度      | ≥56HRC |



| 序号 | 技 术 要 求                             | 备 注          |
|----|-------------------------------------|--------------|
|    | 磁力探伤：<br>按图检查外圆A表面应无裂纹<br>按MT-986检查 | 1. 外圆A保持光洁锐边 |

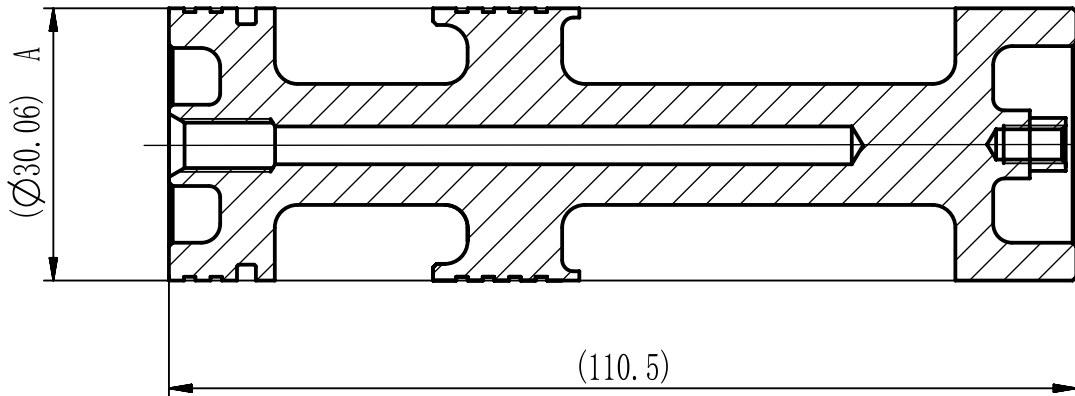
初样研制

会 签

|           |    |      |     |    |      |     |     |               |
|-----------|----|------|-----|----|------|-----|-----|---------------|
| 王普照       |    |      |     |    |      | 编制  | 段长超 | 2023-9-25     |
| 2023-9-27 |    |      |     |    |      | 校对  | 杨一峰 | 2023-9-26     |
|           |    |      |     |    |      | 标审  | 杨科  | 2023-9-27     |
|           |    |      |     |    |      | 审核  |     |               |
| 工艺版次      |    |      |     |    |      | 审定  |     |               |
| A         | 标记 | 更改单号 | 签 名 | 标记 | 更改单号 | 签 名 | 批准  | 陈云维 2023-10-9 |

工艺  
PDM 系统有效文件

|            |          |         |               |       |        |      |          |        |
|------------|----------|---------|---------------|-------|--------|------|----------|--------|
| 一四三厂       |          | 协 作 卡 片 |               |       |        | 编 号  | 69Y-3076 |        |
| 军品机加分厂     |          |         |               |       |        | 工序号  | 70       |        |
| 产品型号       | J/RT-148 | 零组件号    | J/RT148-02-02 | 零组件名称 | 1区计量活门 |      | 共 1 页    | 第 1 页  |
| 工 序<br>名 称 | 烧伤检查     |         |               | 毛坯类型  | 棒料     | 协作单位 | 质量部      |        |
|            |          |         |               | 材料牌号  | 9Cr18  |      | 硬度       | ≥56HRC |



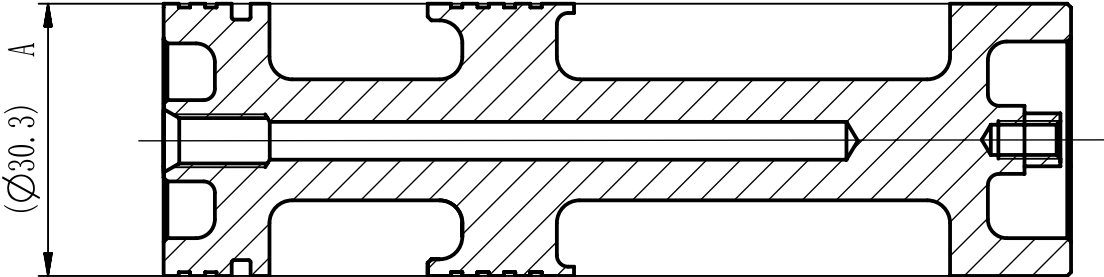
| 序号 | 技 术 要 求                             | 备 注          |
|----|-------------------------------------|--------------|
|    | 磁力探伤：<br>按图检查外圆A表面应无烧伤<br>按MT-986检查 | 1. 表面A保持光洁锐边 |

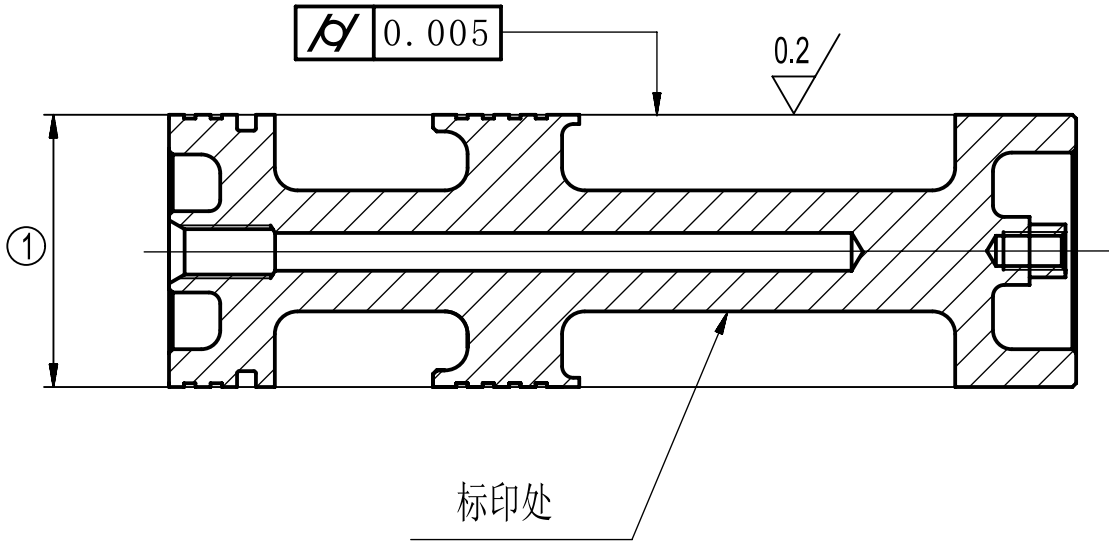
初样研制

会 签

|           |    |      |     |    |      |     |     |               |
|-----------|----|------|-----|----|------|-----|-----|---------------|
| 王普照       |    |      |     |    |      | 编制  | 段长超 | 2023-9-25     |
| 2023-9-27 |    |      |     |    |      | 校对  | 杨一峰 | 2023-9-26     |
|           |    |      |     |    |      | 标审  | 杨科  | 2023-9-27     |
|           |    |      |     |    |      | 审核  |     |               |
| 工艺版次      |    |      |     |    |      | 审定  |     |               |
| A         | 标记 | 更改单号 | 签 名 | 标记 | 更改单号 | 签 名 | 批准  | 陈云维 2023-10-9 |

工艺  
PDM 系统有效文件

|                                                                                               |               |         |         |               |         |                   |               |               |       |       |
|-----------------------------------------------------------------------------------------------|---------------|---------|---------|---------------|---------|-------------------|---------------|---------------|-------|-------|
| 一四三厂                                                                                          |               | 工 序 卡 片 |         |               |         |                   | 编 号           | 69Y-3076      |       |       |
| 军品机加分厂                                                                                        |               |         |         |               |         |                   | 工 序 号         | 75            |       |       |
| 产 品 型 号                                                                                       | J/RT-148      |         | 零 组 件 号 | J/RT148-02-02 |         | 零 组 件 名 称         | 1区计量活门        |               | 共 1 页 | 第 1 页 |
| 毛 坯 类 型                                                                                       | 棒 料           | 硬 度     | ≥56HRC  |               | 设 备 名 称 | 设 备 型 号           | 程 序 编 号       |               |       |       |
| 材 料 牌 号                                                                                       | 9Cr18         |         |         |               | 清 洗 槽   |                   |               |               |       |       |
| 工 序<br>名 称                                                                                    | 清 洗           |         |         |               |         |                   |               |               |       |       |
|                                                                                               |               |         |         |               |         |                   |               |               |       |       |
| <div></div> |               |         |         |               |         |                   |               |               |       |       |
| 工 步                                                                                           | 加 工 内 容       | 夹 具 图 号 | 刀 具 图 号 | 量 具           |         | 技 术 要 求           |               |               |       |       |
|                                                                                               |               |         |         | 序 号           | 量 具 图 号 |                   |               |               |       |       |
| 1                                                                                             | 用毛刷在溶剂清洗剂中将零件 |         |         |               |         | 1. 外圆A保持光洁<br>锐边。 |               |               |       |       |
|                                                                                               | 内外表面的附着物涮洗干   |         |         |               |         |                   |               |               |       |       |
|                                                                                               | 净, 并放入干净的零件专  |         |         |               |         |                   |               |               |       |       |
|                                                                                               | 用盒。           |         |         |               |         |                   |               |               |       |       |
|                                                                                               |               |         |         |               |         |                   |               |               |       |       |
| 初 样 研 制                                                                                       |               |         |         |               |         |                   |               |               |       |       |
|                                                                                               |               |         |         |               |         |                   |               |               |       |       |
| 会 签                                                                                           |               |         |         |               |         |                   |               |               |       |       |
|                                                                                               |               |         |         |               |         | 编制                | 段长超 2023-9-25 |               |       |       |
|                                                                                               |               |         |         |               |         | 校对                | 杨一峰 2023-9-26 |               |       |       |
|                                                                                               |               |         |         |               |         | 标审                | 杨科 2023-9-27  |               |       |       |
|                                                                                               |               |         |         |               |         | 审核                |               |               |       |       |
| 工 艺 版 次                                                                                       |               |         |         |               |         | 审定                |               |               |       |       |
| A                                                                                             | 标记            | 更改单号    | 签 名     | 标记            | 更改单号    | 签 名               | 批准            | 陈云维 2023-10-9 |       |       |



初样研制

会 签

吴昊

2023-9-27

工艺版次

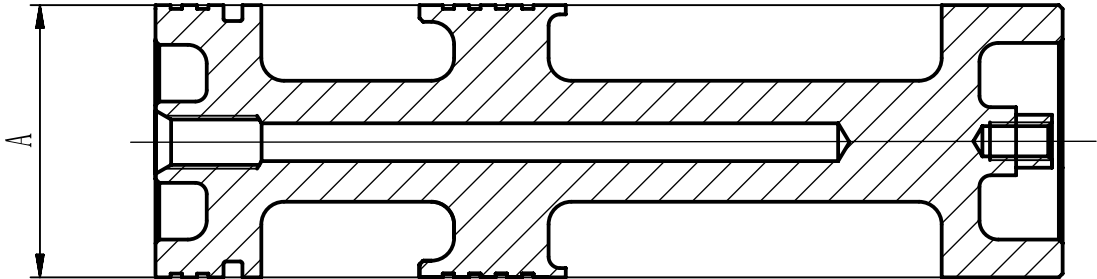
A

|         |                                      |             |               |          |        |     |               |          |         |
|---------|--------------------------------------|-------------|---------------|----------|--------|-----|---------------|----------|---------|
| 一 四 三 厂 |                                      | 检 验 工 序 卡 片 |               |          |        |     | 编 号           | 69Y-3076 |         |
| 军品机加分厂  |                                      |             |               |          |        |     | 工 序 号         | 80       |         |
| 产品型号    | J/RT-148                             | 零组件号        | J/RT148-02-02 | 零组件名称    | 1区计量活门 |     | 共 1 页         | 第 1 页    |         |
| 材料牌号    | 9Cr18                                |             |               | 毛坯类型     | 棒料     | 硬 度 | ≥56HRC        |          |         |
| 编号      | 检 验 内 容                              |             |               | 量 具 图 号  |        | 编号  | 检 验 内 容       |          | 量 具 图 号 |
| I       | 外观检查                                 |             |               |          |        | V   | 开出质量保证单       |          |         |
| 1       | 检查零件应无碰伤, 压                          |             |               |          |        |     |               |          |         |
|         | 伤, 划伤, 锈蚀等缺陷                         |             |               |          |        |     |               |          |         |
| 2       | 检查零件表面粗糙度                            |             |               | 粗糙度比较样块  |        |     |               |          |         |
| 3       | 用冷光源检查零件表面应无毛刺                       |             |               |          |        |     |               |          |         |
|         |                                      |             |               |          |        |     |               |          |         |
| II      | 尺寸检查                                 |             |               |          |        |     |               |          |         |
| ①       | Ø30.06 <sup>0</sup> <sub>-0.02</sub> |             |               | 块规       |        |     |               |          |         |
|         |                                      |             |               | 指示尺0.001 |        |     |               |          |         |
|         |                                      |             |               |          |        |     |               |          |         |
|         |                                      |             |               |          |        |     |               |          |         |
|         |                                      |             |               |          |        |     |               |          |         |
| III     | 形位公差检查                               |             |               | 块规       |        |     |               |          |         |
|         | 0.005                                |             |               | 指示尺0.001 |        |     |               |          |         |
|         |                                      |             |               |          |        |     |               |          |         |
|         |                                      |             |               |          |        |     |               |          |         |
| IV      | 标印                                   |             |               |          |        |     |               |          |         |
|         | 在标印处用激光标印机                           |             |               |          |        |     |               |          |         |
|         | 标印零件批次号                              |             |               |          |        |     |               |          |         |
|         |                                      |             |               |          |        |     |               |          |         |
| 技 术 要 求 | 1.外圆①保持光洁锐边                          |             |               |          |        |     |               |          |         |
|         |                                      |             |               |          |        | 编制  | 段长超 2023-9-25 |          |         |
|         |                                      |             |               |          |        | 校对  | 杨一峰 2023-9-26 |          |         |
|         |                                      |             |               |          |        | 标审  | 杨科 2023-9-27  |          |         |
|         |                                      |             |               |          |        | 审核  |               |          |         |
|         |                                      |             |               |          |        | 审定  |               |          |         |
| 标记      | 更改单号                                 | 签 名         | 标记            | 更改单号     | 签 名    | 批准  | 陈云维 2023-10-9 |          |         |

艺

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|            |               |         |        |               |      |       |               |     |          |       |
|------------|---------------|---------|--------|---------------|------|-------|---------------|-----|----------|-------|
| 一 四 三 厂    |               | 工 序 卡 片 |        |               |      |       |               | 编 号 | 69Y-3076 |       |
| 军品机加分厂     |               |         |        |               |      |       |               | 工序号 | 85       |       |
| 产品型号       | J/RT-148      |         | 零组件号   | J/RT148-02-02 |      | 零组件名称 | 1区计量活门        |     | 共 1 页    | 第 1 页 |
| 毛坯类型       | 棒料            | 硬度      | ≥56HRC |               | 设备名称 | 设备型号  | 程序编号          |     |          |       |
| 材料牌号       | 9Cr18         |         |        |               |      |       |               |     |          |       |
| 工 序<br>名 称 | 包 装           |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
| 工步         | 加 工 内 容       | 夹具图号    | 刀具图号   | 量 具           |      |       |               |     |          |       |
|            |               |         |        | 序 号           | 量具图号 |       |               |     |          |       |
| 1          | 零件表面刷上防锈油后，   |         |        |               |      |       |               |     |          |       |
|            | 用油封纸包裹放入干净零   |         |        |               |      |       |               |     |          |       |
|            | 件盒内           |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      |       |               |     |          |       |
| 技 术<br>要 求 | 1. 外圆A保持光洁锐边。 |         |        |               |      |       |               |     |          |       |
|            |               |         |        |               |      | 编制    | 段长超 2023-9-25 |     |          |       |
|            |               |         |        |               |      | 校对    | 杨一峰 2023-9-26 |     |          |       |
|            |               |         |        |               |      | 标审    | 杨科 2023-9-27  |     |          |       |
|            |               |         |        |               |      | 审核    |               |     |          |       |
|            |               |         |        |               |      | 审定    |               |     |          |       |
| 标记         | 更改单号          | 签 名     | 标记     | 更改单号          | 签 名  | 批准    | 陈云维 2023-10-9 |     |          |       |